



SESSION 02

STATE FEEDBACK, LQR, AND OBSERVERS

Designing feedback gains, optimal regulators, and state estimators.

ACTIVE TRACK

CONTROL + ESTIMATION



CONTROL NEEDS KNOWLEDGE

Why this matters

An autonomous robot must choose actions from information it can actually trust.

CONTROL QUESTION

**Given the state x ,
what input u should
the robot apply?**



ESTIMATION QUESTION

**Given sensor output y ,
what state $\hat{x}$ should
the robot believe?**

Feedback acts on state. Observers make hidden state usable.



FROM FOUNDATION TO DESIGN

The Lecture 01 bridge

Plant:

$$\dot{x} = A x + B u$$

$$y = C x + D u$$

Feedback:

$$u = -K x$$

Closed loop:

$$\dot{x} = (A - B K) x$$

- Lecture 01 defined the state-space language.
- Lecture 02 designs K and estimates x when sensors are incomplete.
- The whole autonomy loop becomes control plus estimation.

This is the first real controller-design layer.



WHAT MUST K ACHIEVE?

Closed-loop stability

K is useful only if the closed-loop dynamics behave

CONTINUOUS TIME

$A_{cl} = A - B K$
stable if $\text{Re}(\lambda_i(A_{cl})) < 0$

DISCRETE TIME

$x[k+1] = A_d x[k] + B_d u[k]$
stable if $|\lambda_i(A_d - B_d K)| < 1$

ROBOTICS TRANSLATION

errors should decay
motion should not explode
actuation should remain feasible

Eigenvalues are not decoration; they are the time-behavior fingerprints.



WHAT ARE WE ASSUMING?

State feedback assumptions

Full-state feedback is a clean mathematical starting point, not the whole robot.

- The state x is available or estimated accurately enough.
- The pair (A, B) is controllable, or at least stabilizable.
- The model is locally valid around the operating point.
- Actuator saturation, delays, and constraints are ignored at first.
- The goal is regulation unless we explicitly define reference tracking.

If x is not available, $u = -Kx$ is a fantasy wearing a lab coat.



DIRECT EIGENVALUE DESIGN

Pole placement

Choose desired closed-loop poles:
 λ_{desired}

Find K such that:
 $\text{eig}(A - B K) = \lambda_{\text{desired}}$

Requirement:
 $\text{rank}([B \ AB \ \dots \ A^{(n-1)}B]) = n$

- For controllable systems, K can place closed-loop poles.
- Farther-left continuous poles usually mean faster response.
- Aggressive poles demand more input and amplify modeling errors.
- Pole placement gives control authority, not automatic tests.

It answers where the poles go, but not what tradeoff is best.



SAME 1D MODEL, NOW CONTROLLED

Marine robot feedback example

$$\mathbf{x} = \begin{bmatrix} p \\ v \end{bmatrix}$$

$$\dot{p} = v$$

$$\dot{v} = -(b/m)v + (1/m)u$$

$$u = -K\mathbf{x} = -\begin{bmatrix} k_p & k_v \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix}$$

$$\mathbf{A}_{cl} = \mathbf{A} - \mathbf{B} \mathbf{K}$$

$$\mathbf{A}_{cl} = \begin{bmatrix} 0 & 1 \\ -k_p/m & -(b+k_v)/m \end{bmatrix}$$

$$p_{ddot} + ((b+k_v)/m)p_{dot} + (k_p/m)p = 0$$

Position gain creates restoring action; velocity gain adds damping.



REGULATION IS NOT THE WHOLE MISSION

Reference tracking

Regulation:

$$u = -K x$$

drives $x \rightarrow 0$

Tracking:

$$e = x - x_{\text{ref}}$$

$$u = u_{\text{ref}} - K e$$

For constant-speed marine motion:

$$u_{\text{ref}} = b v_{\text{ref}}$$

- LQR and basic state feedback naturally regulate deviations.
- For a mission target, shift the coordinates around a reference.
- Feedforward u_{ref} handles known equilibrium input.
- Feedback handles disturbance and model error

Always ask: are we driving to zero, or driving to a desired mission



CHOOSING K BY COST

Why LQR?

Manual gain tuning becomes ugly when several states and actuators interact.

- Fast response usually requires more control effort.
- Low effort usually means slow correction and larger error.
- Some states matter more than others.
- Some actuators are expensive, weak, or energy-limited.
- LQR turns this into a precise quadratic optimization problem.

LQR is not magic. It is disciplined compromise written as mathematics.



THE COST FUNCTION

LQR objective

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

$$Q = Q^T \succeq 0 \quad \text{state penalty}$$

$$R = R^T \succ 0 \quad \text{input penalty}$$

- Minimize accumulated state error and control effort.
- Q says which state deviations hurt.
- R says how expensive actuation is.
- For continuous-time infinite-horizon LQR, the system must be stabilizable and the unpenalized unstable modes must be detectable through Q .

The controller becomes optimal for this model, this cost, and these assumptions.



WHERE THE GAIN COMES FROM

Riccati equation and K

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

$$K = R^{-1} B^T P$$

$$u = -K x$$

- P is the stabilizing solution of the continuous algebraic Riccati equation.
- K is not guessed; it is computed from A, B, Q, and R.
- If the assumptions fail, a stabilizing LQR gain may not exist.
- The closed-loop matrix is still $A_{cl} = A - B K$.

Riccati is the engine room. Q and R are the control-room knobs.



STATE PAIN VS INPUT PAIN

Meaning of Q and R

HIGHER Q

**More state
correction**

HIGHER R

**Less actuator
aggression**

RELATIVE SCALE

**Units matter
a lot**

```
Q = diag(q_position, q_velocity, ...)  
R = diag(r_thruster_1, r_thruster_2, ...)
```

Large q_position -> position error is expensive

Large r_thruster -> thrust is expensive

Bad scaling creates nonsense. Normalize units before worshipping gains.



1D POSITION-VELOCITY REGULATOR

Marine LQR setup

```
x = [ position_error ; velocity_error ]  
u = thrust correction
```

```
Q = [ q_p  0 ]  
    [ 0   q_v ]
```

```
R = [ r ]
```

```
J = integral (q_p p^2 + q_v v^2 + r u^2) dt
```

Solve CARE -> P

Compute $K = R^{-1} B^T P$

Apply: $u = u_{ref} - K(x - x_{ref})$

TUNING INTUITION

q_p up -> less position error, more thrust. r up -> smoother, slower, lower energy.

Higher q_p pulls position error down faster. Higher r conserves thrust.



WHAT LQR DOES NOT SOLVE

LQR limitations

LQR is elegant, but the ocean does not read your Riccati equation.

- It assumes a linear time-invariant model, usually local around an operating point.
- It assumes quadratic cost and unconstrained input.
- Actuator saturation breaks the optimality claim and can destabilize naive designs.
- Large nonlinear excursions may leave the valid region of the linearized model.
- It needs state feedback; incomplete sensing requires an observer or filter.

Use LQR as a strong baseline, not a religion.



CONTROL NEEDS STATE

Why estimation enters

Robots rarely measure every useful state directly.

SENSORS GIVE

position
depth
heading
range
bearing

CONTROL MAY NEED

velocity
bias terms
current estimate
unmeasured modes
filtered pose



State estimation builds $\hat{x}$ from model prediction plus sensor correction.



PREDICT + CORRECT

Observer structure



$$\begin{aligned} \dot{x}_{\text{hat}} &= A x_{\text{hat}} + B u + L(y - y_{\text{hat}}) \\ y_{\text{hat}} &= C x_{\text{hat}} + D u \end{aligned}$$

The innovation $y - y_{\text{hat}}$ is the sensor disagreement that corrects the model.



DETERMINISTIC STATE RECONSTRUCTION

Luenberger observer

$$\begin{aligned} \dot{x}_{\hat{}} &= A x_{\hat{}} + B u + L (y - y_{\hat{}}) \\ y_{\hat{}} &= C x_{\hat{}} + D u \end{aligned}$$

$$\begin{aligned} e &= x - x_{\hat{}} \\ \dot{e} &= (A - L C) e \end{aligned}$$

- L is the observer gain.
- If (A, C) is observable, L can place observer error poles.
- If only detectable, unobservable modes must already be stable.
- The observer should converge faster than the controller, but not so fast that it injects sensor noise violently.

K controls the plant. L controls belief error.



HOW FAST SHOULD ESTIMATION CONVERGE?

Observer pole design

Controller poles:
 $\text{eig}(A - B K)$

Observer poles:
 $\text{eig}(A - L C)$

Rule of thumb:
observer dynamics often 2x to 5x faster

- Faster observer poles reduce estimation lag.
- Too-fast observers amplify measurement noise and discretization artifacts.
- Poor C matrices leave some state directions invisible.
- Observer design is limited by observability, sensor quality, and model quality.

A nervous observer can make a calm controller look possessed.



CONTROL WITH ESTIMATED STATE

Observer-based feedback

$$u = -K \hat{x}$$

$$\text{Plant:} \quad \dot{x} = A x + B u$$

$$\text{Observer:} \quad \dot{\hat{x}} = A \hat{x} + B u + L(y - \hat{y})$$

Closed-loop modes:
 $\text{eig}(A-BK) \cup \text{eig}(A-LC)$

- For deterministic LTI systems under standard assumptions, control and observer gains can be designed separately.
- This is the separation principle.
- The controller stabilizes the plant state.
- The observer stabilizes estimation error.

One loop acts on the robot. The other loop cleans the robot's self-image.



WHEN SENSORS ARE NOISY

Noise-aware estimation

$$\begin{aligned} \dot{x} &= A x + B u + w \\ y &= C x + v \end{aligned}$$

w = process/model disturbance

v = measurement noise

- A Luenberger observer uses a designed fixed gain L .
- A Kalman filter chooses L from noise covariance assumptions.
- With linear dynamics, Gaussian noise, and correct covariance models, the Kalman filter minimizes expected squared estimation error.
- Bad noise assumptions still produce confident wrongness. Tiny submarine ghost with a clipboard says: beware.

Kalman filtering is the statistical successor to observers.



ESTIMATING VELOCITY FROM POSITION

Marine observer example

Plant:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \end{bmatrix}$$

$$\mathbf{y} = \mathbf{p}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -b/m \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\mathbf{Omat} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$\mathbf{CA} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\text{rank}(\mathbf{Omat}) = 2$$

=> observable

Observer:

$$\dot{\mathbf{p}}_{\text{hat}} = \mathbf{v}_{\text{hat}} + \mathbf{L}_1(\mathbf{p} - \mathbf{p}_{\text{hat}})$$

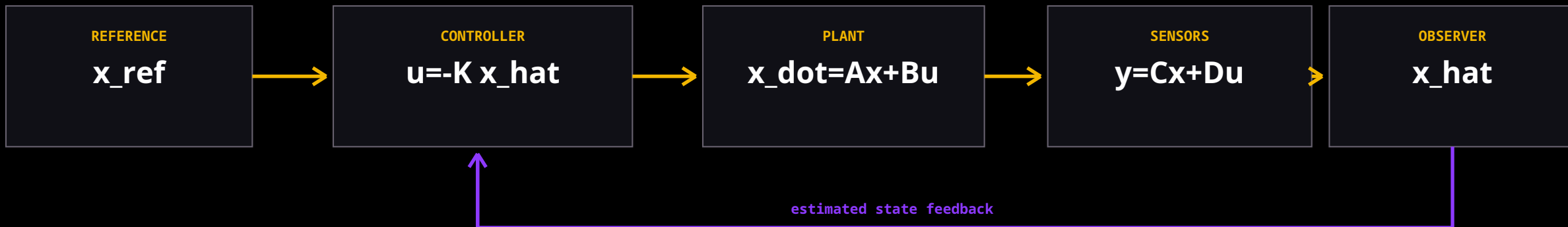
$$\dot{\mathbf{v}}_{\text{hat}} = -(b/m)\mathbf{v}_{\text{hat}} + (1/m)\mathbf{u} + \mathbf{L}_2(\mathbf{p} - \mathbf{p}_{\text{hat}})$$

Position measurement plus dynamics can reveal velocity over time.



CONTROL + ESTIMATION

Full autonomy loop



Local estimation and feedback are the unit cell. Swarms repeat the cell across many agents.



WHAT SHOULD SURVIVE?

KEY TAKEAWAYS

- State feedback computes action from state: $u = -Kx$.
- Closed-loop behavior is governed by $A - BK$.
- LQR chooses K by minimizing state error and input effort.
- Observers reconstruct hidden state through model prediction and sensor correction.
- Observer-based feedback uses $u = -K \hat{x}$.
- Control without estimation is blind. Estimation without control is commentary.

NEXT: MPC, constraints, Kalman filtering depth, and multi-agent consensus.

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